INPUT

```
data("BJsales")
print(BJsales)
plot(BJsales,main="BJsales Time Series",ylab="Sales",xlab="Month",col="orange")
time<-1:length(BJsales)
trend_fit<-lm(BJsales~poly(time,2))
plot(BJsales,main="BJsales with Polynomial Trend",ylab="Sales",xlab="Month",col="orange")
lines(time,predict(trend_fit),col="green",lwd=2)
residuals<-BJsales-predict(trend_fit)
```

#D

```
diff_BJsales<-diff(BJsales)
plot(diff_BJsales,main="Differenced BJsales",ylab="Differenced Sales",xlab="Month",col="blue")
library(tseries)
adf.test(diff_BJsales)
```

#E

```
par(mfrow=c(1,2))
acf(diff_BJsales,main="ACF of Differenced BJsales")
pacf(diff_BJsales,main="PACF of Differenced BJsales")
```

#F

```
library(forecast)
fit<-auto.arima(BJsales)
summary(fit)
```

#G

```
residuals<-residuals(fit)
par(mfrow=c(1,2))
plot(residuals,main="Residuals of Fitted Model",ylab="Residuals",xlab="Month",col="blue")
acf(residuals,main="ACF of Residuals")
```

#H

```
forecast_values<-forecast(fit,h=10)
plot(forecast_values,main="10-step Forcast for BJsales",ylab="Sales",xlab="Month")
```

#i

```
plot(BJsales,main="Actual vs Predicted BJsales",ylab="Sales",xlab="Month",col="blue")
lines(forecast_values$mean,col="red",lwd=2)
```

#2

```
data("co2")
print(co2)
```

#b

```
plot(co2,main="co2 Time Series",ylab="Month",xlab="Year",col="purple")
```

#c

```
diff_trend<-diff(co2,differences = 1)
diff_seasonal<-diff(diff_trend,lag=12)
plot(diff_seasonal,main="Differenced Series(Stationary",ylab="Differenced
CO2",xlab="Time",col="green",type="o")
library(tseries)
adf.test(diff_seasonal)
```

#d

```
par(mfrow=c(1,2))
acf(diff_seasonal,main="ACF of Stationary Series")
pacf(diff_seasonal,main="PACF of Stationary Series")
```

#e

```
library(forecast)
fit<-auto.arima(co2)
summary(fit)
```

#f

```
residuals<-residuals(fit)
par(mfrow=c(1,2))
plot(residuals,main="Residuals of Fitted Model",ylab="Month",xlab="Year",col="blue")
acf(residuals,main="ACF of Residuals")
```

#g

```
forecast_values<-forecast(fit,h=12)
plot(forecast_values,main="12-step Forcast for co2",ylab="Month",xlab="Year")
```

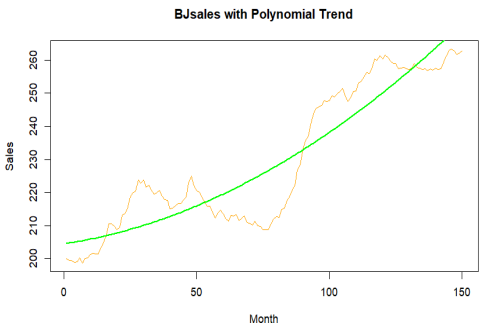
#h

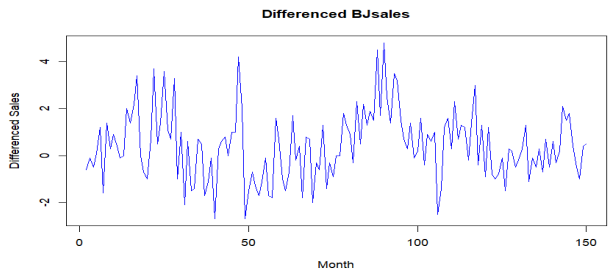
```
plot(co2,main="Actual vs Predicted co2",ylab="Month",xlab="Year",col="purple")
lines(forecast_values$mean,col="red",lwd=2)
```

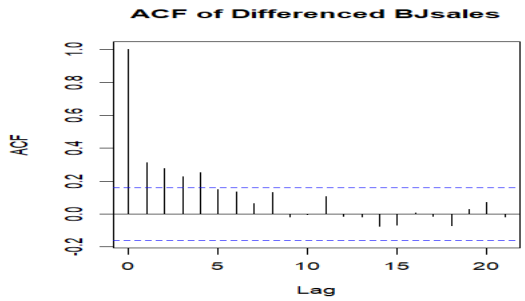
OUTPUT

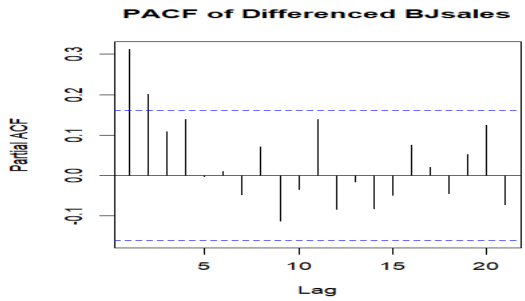
#1

```
print(BJsales)
Time Series:
Start = 1
End = 150
Frequency = 1
  [1] 200.1 199.5 199.4 198.9 199.0 200.2 198.6 200.0 200.3 201.2 201.6 201.5 201.5 203.5 204.9
207.1 210.5
……..
[120] 260.4 261.6 260.8 259.8 259.0 258.9 257.4 257.7 257.9 257.4 257.3 257.6 258.9 257.8 257.7
257.2 257.5
[137] 256.8 257.5 257.0 257.6 257.3 257.5 259.6 261.1 262.9 263.3 262.8 261.8 262.2 262.7
```









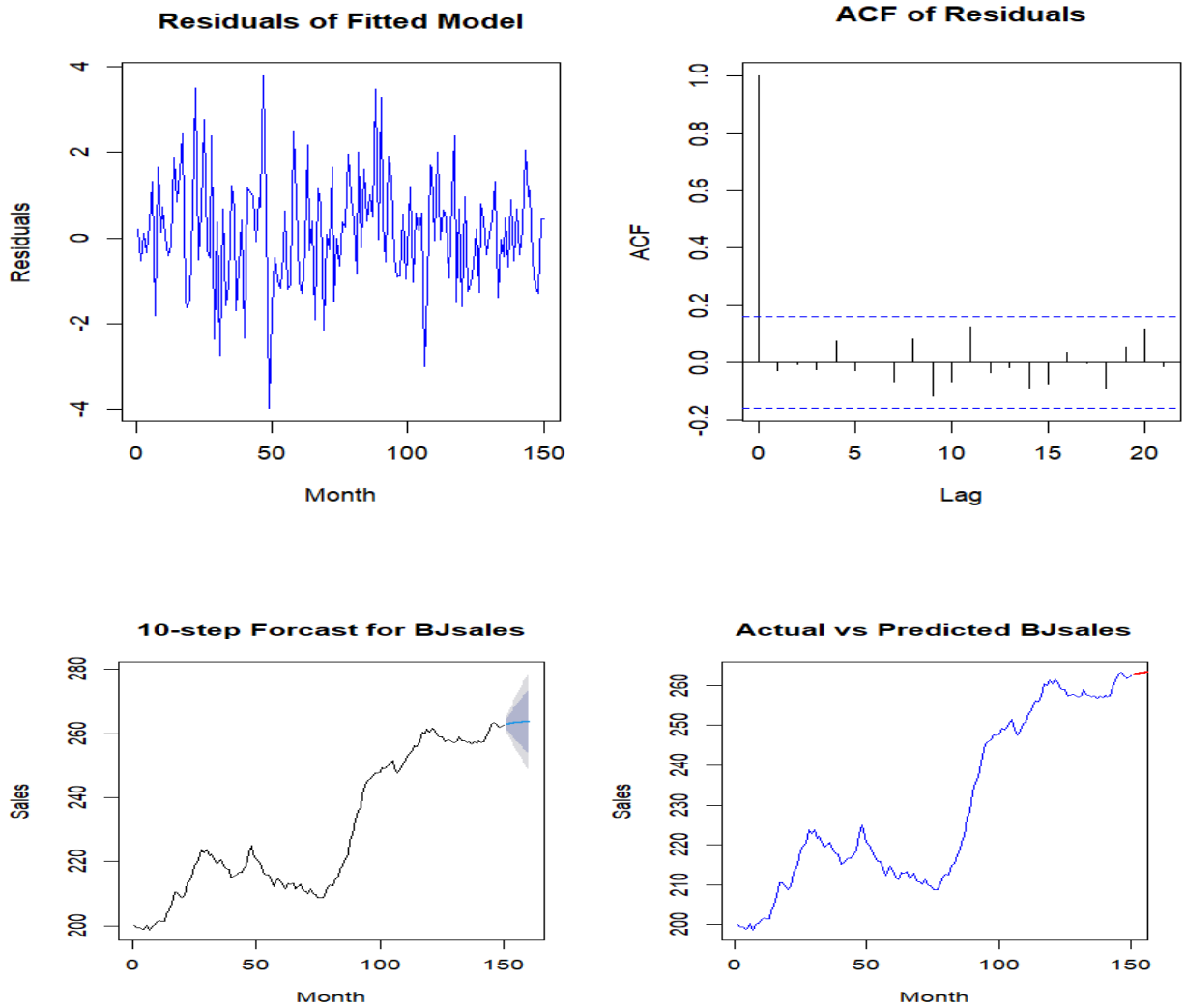

```
>summary(fit)
Series: BJsales
```

ARIMA(1,1,1)

Coefficients:
```
       ar1     ma1
    0.8800  -0.6415
s.e.  0.0644   0.1035
```

sigma^2 = 1.8: log likelihood = -254.37
AIC=514.74 AICc=514.9 BIC=523.75

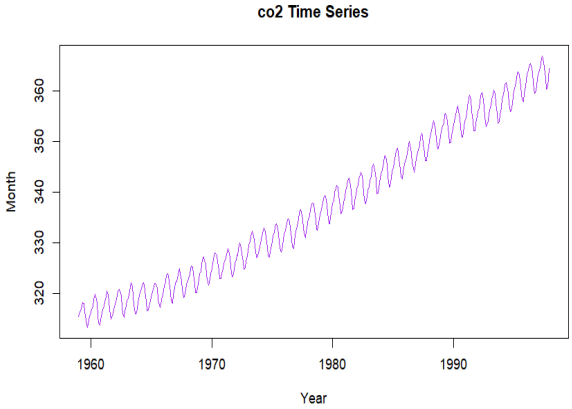
Training set error measures:
```
                    ME     RMSE    MAE       MPE      MAPE      MASE       ACF1
Training set 0.1457572 1.328119 1.0447 0.06512899 0.4600935 0.8997702 -0.02622396
```

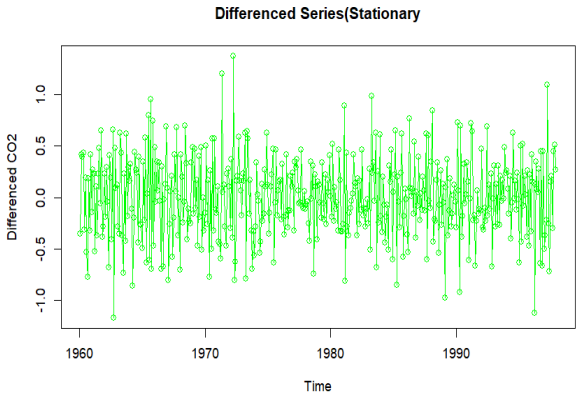


#2

print(co2)
```
      Jan    Feb    Mar    Apr    May    Jun    Jul    Aug    Sep    Oct    Nov    Dec
1959 315.42 316.31 316.50 317.56 318.13 318.00 316.39 314.65 313.68 313.18 314.66 315.43
1960 316.27 316.81 317.42 318.87 319.87 319.43 318.01 315.74 314.00 313.68 314.84 316.03
```

……362.38 1997 363.23 364.06 364.61 366.40 366.84 365.68 364.52 362.57 360.24 360.83 362.49 364.34

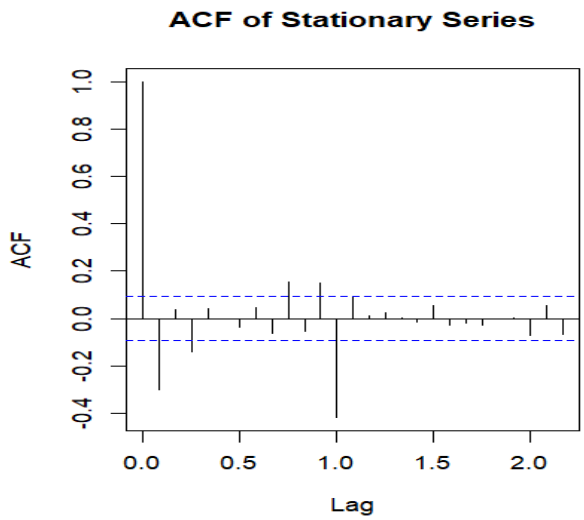


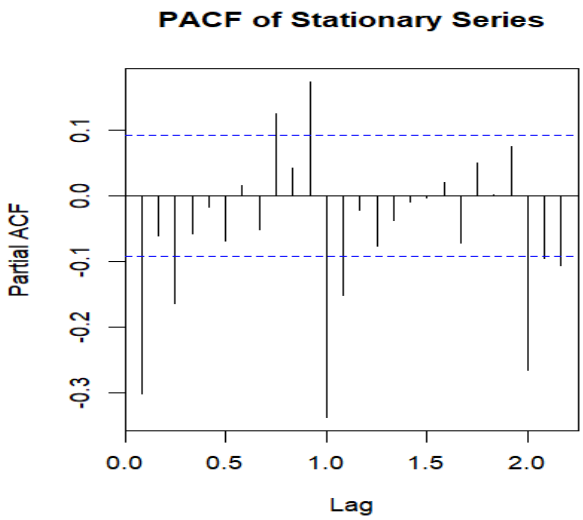


adf.test(diff_seasonal)

Augmented Dickey-Fuller Test

data: diff_seasonal
Dickey-Fuller = -8.9846, Lag order = 7, p-value = 0.01
alternative hypothesis: stationary





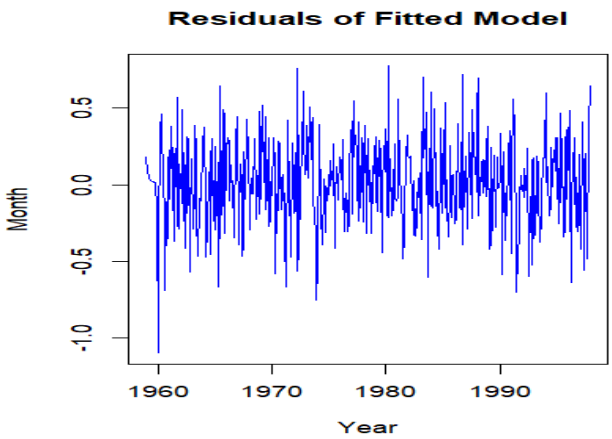
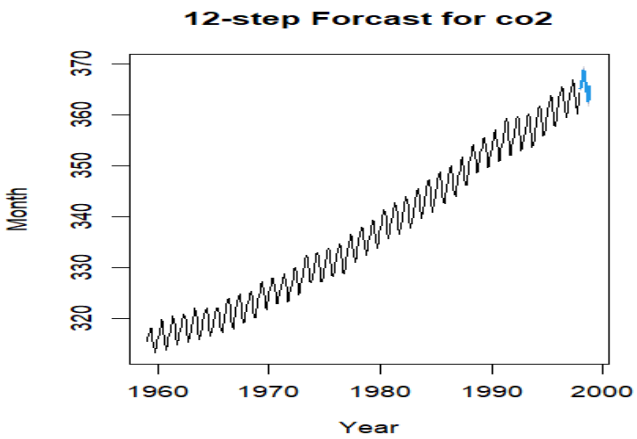
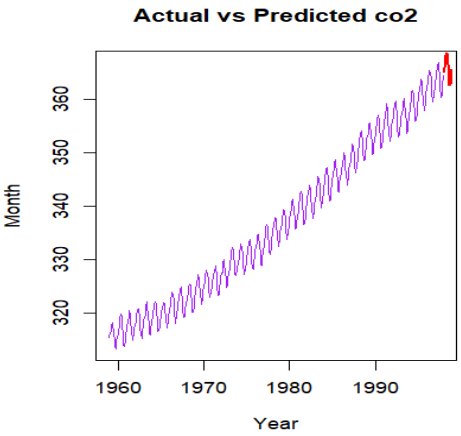
summary(fit)
Series: co2
ARIMA(1,1,1)(1,1,2)[12]

Coefficients:

	ar1	ma1	sar1	sma1	sma2
	0.2569	-0.5847	-0.5489	-0.2620	-0.5123
s.e.	0.1406	0.1204	0.5879	0.5701	0.4819

sigma^2 = 0.08576: log likelihood = -84.39
AIC=180.78 AICc=180.97 BIC=205.5

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1

Training set 0.01742092 0.287159 0.2303994 0.00507377 0.06845665 0.1819636 -0.00285809